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DESIGN RESEARCH ESSAY

Wicked Problems in Architectural Research: The Role of Research by Design

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PhD students working in the field of architectural design often miss the reference to a consistent theoretical framework and a proper system of inquiry while working with the research design of their projects. Simultaneously, some of the architectural research questions and hypotheses are formulated as addressing wicked problems which causes methodological challenges. The aim of the paper is to put research by design approaches in the broader context of systems of inquiry and to emphasize their possible role to examine and resolve wicked problems.

The paper first reviews some of the underlying assumptions of architectural research versus the assumptions behind different research paradigms characteristic for other fields. Then, the approaches and the methods in architectural research relevant to the chosen ontological, epistemological, and axiological assumptions are discussed. The theory of wicked problems is recalled as a relevant one for the complex, uncertain, projective tasks of architectural research. Research by design is seen as a method suitable for the investigation of wicked problems. It can reduce the epistemic uncertainty. Approaches of research by design are becoming more acknowledged in third cycle education in architecture. Two examples of recent PhD theses are briefly discussed in the paper in relation to the wicked aspects of their research/design problems, the research paradigm adopted and the role of research by design methodology. More conscious delineations and elaborations of a proper system of inquiry and the conscious use of research by design methodology can help in developing academic rigor of research studies addressing complex societal challenges.

Keywords: research paradigm; architectural research; research by design; wicked problems; epistemic uncertainty; doctoral studies

Introduction

As the director of a Graduate School of Architecture, I have observed that the PhD students often miss the reference to theory and a proper system of inquiry (also a worldview or paradigm) while working with the design of research for their projects. Simultaneously, I have realized that some of the architectural research questions and hypotheses are formulated as addressing wicked problems which causes methodological challenges, not least for more inexperienced researchers. However, to argue for the application of a proper methodology, an ontological and epistemological perspective has to be chosen, which then also calls for an awareness of the wider epistemological landscapes.

The theory of 'wicked problems', in contrast to tame problems typical for 'scientific' disciplines, was proposed by Horst Rittel & Melvin Webber in 1973 [1]. They established ten criteria describing such problems, which according to Robert Farrell and Cliff Hooker [2] can be reduced to their three main characteristics, common to both science and design: agent finitude, understood as capacity limitations; complexity, understood as uncertainty and irreversibility embedded in 'complex-systems nature of the world'; and normativity, referring to human values and norms. Solutions to wicked problems are not judged by agreed criteria as being correct or not (i.e. true or false) but rather as good or bad according to applied norms. Wicked problems often come from practice and are thus expected to be solved by practice, in contrast to issues in basic sciences which in general are resolved within a theoretical space under ideal conditions.

Within the applied sciences (architecture and engineering included), the constructed models of reality are uncertain because of lack of sufficient knowledge, the introduction of simplifications, and by not taking the variability of the main characteristics fully into account. As a result, the behaviour of the application in practice is difficult to predict. The concept of 'model' introduces a kind of wickedness to the issue analysed. However, in many engineering applications the wickedness is not crucial and can be successfully addressed by the reductionistic modelling strategies or analytical approaches to deal with uncertainties. Architectural research, on the other hand, often deals with very complex problems based in specific situations where human values and norms play an important role. The proper modelling approach refers mostly to the soft system's representations describing fuzzy situations involving cultural and social considerations and typically wicked problems offer the possibility of an enumerable set of potential solutions. It results in higher epistemic uncertainty (i.e. uncertainty in terms of producing scientific knowledge) in the research, and lower academic rigour.

The aim of this paper is to emphasize the possible role of research by design as one of the methodologies to investigate, examine and resolve wicked problems. Any research methodology (the process of research) is built alongside ontological and epistemological assumptions – it means, assumptions about what reality is, and how we know reality, as well as what appropriate methods and tools to use to acquire knowledge. In this essay, different research paradigms in architectural research will be described and exemplified, and the concept of wicked problems in relation to research by design methodology is introduced. Two relevant PhD theses, by Joanna Gregorowicz-Kipszak [3] and Nils Björling [4], are also discussed (based upon short interviews with the authors) in regard to the wicked aspects of the research/design problems involved, the research paradigms adopted, and the role of research by design methodology.

Architectural research: philosophical underpinnings and research paradigms

The goal of research is to create new knowledge, and from that point of view, it is important for any researcher to know about different forms of knowledge as well as how and in what contexts they are created. The Swedish philosopher Bernt Gustavsson has been a frequent lecturer in the Swedish Research School in Architecture (ResArc), and in that context he wrote an essay discussing the many faces of knowledge [5], based upon his book titled *Three Forms of Knowledge in a Historical Perspective* [6]. In his essay, the three forms of knowledge formulated by Aristotle are given a contemporary explanation: knowledge as ***episteme*** (theoretical-scientific); knowledge as ***techne*** (practical-productive, including technique, craft and art, and connected with tacit knowledge); and knowledge as ***phronesis*** (practical wisdom, ethically rooted, as an interpretive form of understanding). They refer to the original Aristotelian concepts of *'theoria'* (philosophical activity), *'poiesis'* (creating products), and '*praxis'* (human actions to make a good life). In Gustavsson's reading, the focus of the basic sciences is concentrated on ***episteme*** while the applied sciences include the other kinds of knowledge, ***techne*** and ***phronesis***. In this context, architectural research aspires to belong to ***episteme***, but often falls into ***techne*** and ***phronesis***. Especially important for architects is tacit knowledge, the personal knowledge which is difficult to communicate verbally but can be a part of a creative discovery process situated in a specific context. It constitutes the pre-phase of knowing in a more explicit way.

Two important philosophers, Karl Popper [7] and Michael Polanyi [8], have greatly contributed to the understanding of different kinds of knowledge. The epistemology of Popper is certainly more applicable for engineers and basic scientists, whereas Polanyi's theory of tacit knowledge can be significant and useful for architecture and art (and the humanities) given that tacit knowledge is embodied in artworks and artistic processes. The artistic view of architectural design, in not being objective, is difficult to evaluate or measure from a scientific point of view but can on the contrary be valued from an individual-, group-, trend- or time-dependent perspective. It is different from scientific knowledge, which according to Popper is produced by refutation or validation of theories, yet what is interesting in Popper's thinking also from this perspective is the acknowledgment of uncertainty as an immanent characteristic of every theory or concept. It is always a challenge how to deal with uncertainty, and it depends on the ontological and epistemological assumptions of the issue being investigated. Not least in cases of practical wicked problems, it is difficult to reach a clear yes or no (deterministic) decision, given the major uncertainty that is often present, and Popper addresses these challenges in an interesting way that is of relevance for all kinds of research.

A research paradigm, or worldview, can best be described as 'the set of common beliefs and agreements shared between scientists about how problems should be understood and addressed' [9] and as such it

can be shown as a hierarchical system of inquiry (**Figure 1**) which is modified from Groat & Wang [10]. A philosophical (ontological, epistemological, and axiological) perspective of a research problem guides the researcher through the variety of methodological approaches, methods and tools. As has also been said: 'Methods and techniques that are at the disposal of the researcher should be subordinated to consideration of purpose and philosophy' [11].

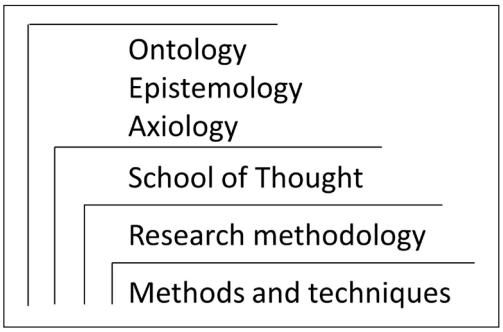


Figure 1: System of inquiry with axiology added, as modified from Groat & Wang [Diagram by the author].

Ontology is concerned with the nature of the investigated world, the objects, and the relations between them. A specific ontology generates specific theories about what can be known and in which way the knowledge is produced and whether something is real. Architectural ontologies can be described as 'the nature of architectural objects and their properties and types, the relations of architectural parts and wholes, and the prospect of architectural causality', and architectural objects are here understood as 'built structures, models, sketches, and plans', but also as 'parts of buildings and buildings considered in context, among other buildings and in landscapes' [12]. Theories of architecture change over time according to the changing perspective on space, form, etc., and the context of various cultural, social, economic conditions should be accounted for. Depending on the philosophical perspective, the objects and the relationship between them can be seen as a simple system or as a complex system depending on which aspects and the character of the relationships among them that are being taken into account. The ontology of a simple system includes small numbers of elements, highly organized interaction, with weak connection to the environment. In contrast, a complex system is characterized by variety, emergence, connectedness and disorder, uncertainty in behaviour, and is largely open to the social and cultural environment.

Epistemology refers to the nature and the forms of knowledge, the possible sources of knowledge, how knowledge can be acquired and communicated. The most common forms of logics are deductive, inductive, and abductive. Deductive inference is typical for natural sciences, whereas inductive reasoning is common in the social sciences and humanities. Abductive thinking is associated with design studies and the search for new ideas. According to Alain Findeli, even some other logics are used in design research, such as phenomenological, narrative, rhetorical, semiotic, hermeneutic, deontic, and heuristic [13].

Axiology, as a study of the nature of values and valuation, could constitute a complementary part of the system of inquiry. In this essay axiology has been added to the philosophical perspective presented by Linda Groat & David Wang [10], since values like well-being, environmental safety, generational equality, social justice, or participation can guide architectural research towards working on wicked problems of sustainable development.

As noted by John Creswell, the worldview for an individual student is shaped by their field of the studies, but also by faculty members and other advisors [14]. It can also reflect the tradition of a graduate school and its curriculum culture [15]. Research can be, but does not have to be, conducted within different schools of thought, with broad theoretical perspectives often related to the multiple disciplines or domains (in cross-disciplinary research, a theoretical framework that explains associated phenomena). It is possible to define a main (central) discipline-specific epistemological perspective for each school of thought [10]. This school of thought then influences how research questions are framed and articulated, and it implies the use of certain methodological approaches in terms of choosing qualitative, quantitative or mixed methods.

Research methodology is an action plan described by a system of methods (quantitative, qualitative or mixed) that lead from research question or hypothesis to knowledge generation, and the chosen methodological approach should always be relevant to the chosen worldview. Architectural research methodologies include for instance case studies, simulations, investigative projects, analyses, interpretations, reflections and argumentations, while the often interdisciplinary, practice-based research fields call for

effective cooperation. Here systems thinking can be used to identify the important variables and their dynamic behaviour, and to facilitate understanding of the system (phenomenon) and the explanation of its emerging properties. For example, Catharina Dyrssen proposes a specific way of architectural thinking as a 'multidimensional mode of construct, perceive and conceptualize complex situations' [16]. Architectural design research can hence be realised by architects-practitioners researching design through their practice, as well as by researchers who are observing and reflecting upon new understandings of design through individual projects, living labs, urban labs and so on, and also by practitioners who use design to investigate and create understanding and knowledge of a situation or problem [17]. So, it is not only about researching into design or creating new understandings of design, but about using design as a method or tool in research. However, architectural research mainly uses integrated methods which are more related to traditions of the humanities, social sciences, technical and natural sciences, and design-based and artistic production of knowledge [18]. Natural science (physics, biology etc) in contrast tends to be about understanding, description and prediction of natural phenomena, while technological science studies the objects designed and constructed by humans [19]. Social science deals with different aspects of human society represented by disciplines like sociology, anthropology, economics etc. Humanities are focused on the product of human society in terms of cultural representations found in art, literature, philosophy and such like.

To generalise, theories tended to be mostly disciplinary whereas practice work is very often interdisciplinary. Architectural research needs cooperation with, but also recognition from, other disciplines and it is important for PhD students in architectural schools to realise that philosophy of science varies among disciplines [20]. Architecture can thus be treated as a discipline of applied science, occasionally using approaches from both social and natural sciences, while design – understood as both product and process – also remains firmly in focus (**Figure 2**).

Generally, research can be initiated from a range of positions and events: it can be from an observation, by a surprising fact, innovative experiment (design), from curiosity, from interesting results in measurements, or indeed by the research question or hypothesis being the result of previous research, not least in the case of design research. Research can often be trigged by the broader challenges in society, calling for practical solutions. As challenges are often not fully understood and practice is complex, context-based, and multidisciplinary, it creates problems in formulating a realistic and empirically researchable question.

A research problem characterized by a research question or a hypothesis is the most important determinant of which research philosophy should be chosen, something which perhaps is already unconsciously chosen or which forms part of the context for the research. The paradigmatic positions based on philosophical

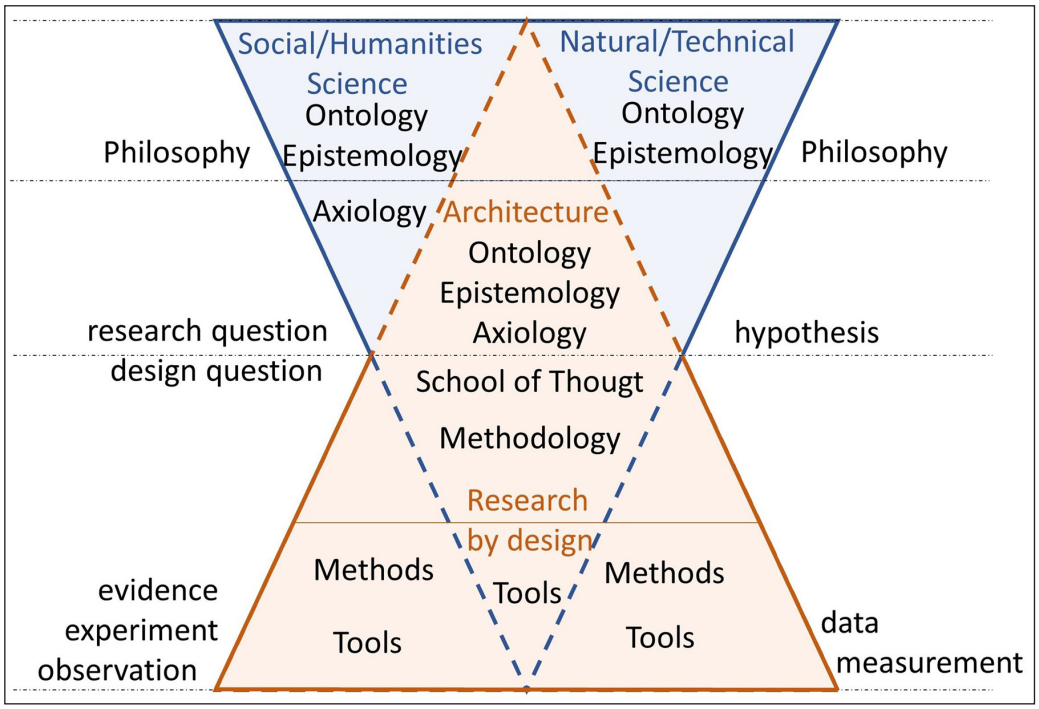


Figure 2: Research paradigms [Diagram by the author].

Worldview	Ontology/Epistemology/Axiology
Positivism	reality is objective and independent of social actors, deterministic, reductionistic, can be known. Theory verifying, value-free.
Post-positivism	reality is objective but can be known only imperfectly and probabilistically. Acknowledging empirical observation and measurement. Theory verifying, value-free.
Pragmatism	not one type of reality, the worldview is chosen to the best way answer research question (multiple perspectives and approaches can be combined), practice oriented.
Critical Realism	reality is objective, stratified (observed, actual, real). Critical inquiry. Interdisciplinary integration of social conditions and norms (values), spatial structures and the action of agents for knowledge production.
Advocacy/ Participatory (Transformative)	reality is objective but can be interpreted – critical inquiry, advocating for the needs of people (participants) that may be marginalized, political, social problem-oriented, with action agenda, collaborative. Research driven by values.
Constructivism	reality is subjective, socially, historically, culturally constructed, pattern recognizing, theory generating, findings emerge through dialogue, linked to values.

Figure 3: List of worldviews/philosophical perspectives [Table by the author].

stance can be identified according to how reality is viewed: objectively, intersubjectively, or subjectively [10]. Architectural research is characterized by a multitude of approaches based on the different traditions and recall different systems of inquiry – and, as such, has 'shifted from positivist, to cognitivist, to discursive, then turning to practice and more recently acknowledging embodied and experimental understanding of the world we all inhibit. Research in all of these forms is still undertaken' [17]. These examples of philosophical perspectives can be grouped into six main worldviews (**Figure 3**), the description of which is based on the literature, and they have here been chosen to exemplify the general worldviews and those discussed in the PhDs analysed.

Wicked problems: theory and methods

Many practical problems, notably those in architecture, require solutions that need to be judged under time pressures by people who have no possibility to predict the consequences of such solutions. Solutions of these kinds cannot be judged as being correct or incorrect. Such problems are called 'wicked problems'. The theory of wicked problems was first introduced by Rittel and Webber [1], and their original definition sets out ten characteristics:

1. There is no definitive formulation of a wicked problem.
2. Wicked problems have no stopping rule.
3. Solutions to wicked problems are not true-or-false, but good-or-bad.
4. There is no immediate and no ultimate test of a solution to a wicked problem.
5. Every solution to a wicked problem is a 'one-shot operation'; because there is no opportunity to learn by trial-and-error, every attempt counts significantly.
6. Wicked problems do not have an enumerable (or an exhaustively describable) set of potential solutions, nor is there a well-described set of permissible operations that may be incorporated into the plan.
7. Every wicked problem is essentially unique.
8. Every wicked problem can be considered to be a symptom of another problem.
9. The existence of a discrepancy representing a wicked problem can be explained in numerous ways and the choice of explanation determines the nature of the problem's resolution.
10. The planner has no right to be wrong.

Many researchers have critically adopted the conceptualisation provided by Rittel and Webber by introducing some other definitions [21]. For example, Farrell & Hooker creatively reformulated the theory to avoid perceived redundancy in the given criteria, as well as to open up the application to a wide range of disciplines where certain strategies to address wicked problems could be beneficial [2]. Every wicked problem is hence treated as unique, and context based, and so Farrell and Hooker proposes these tighter categories instead:

- *Finitude* – understood as limitations on capacity (emergencies of time, cognitive limitations etc.)
- *Complexity* – with uncertainty and irreversibility being embedded in the 'complex-systems nature of the world'
- *Normativity* – usually referred to human values and norms.

Hence, wicked problems are seen as complex, uncertain, confronting value conflicts, involving different stakeholders [22]. Different kinds of uncertainties are relevant to different problems. Uncertainties that arise from natural variability in a system in space and in time (irreducible aleatory uncertainty) can be modelled for example with the help of statistical/probabilistic approaches if enough data is available and if the system's behaviour is predictable [23, 24]. In case of design research, the choice from a plethora of acceptable outcomes relies on the tacit knowledge of the designers concerning intuitions about the historical repertoire, precedents, trends, and future changes.

Planning problems touching social planning situations are inherently wicked in contrast to typical 'tame' problems of science [1]. Farrell and Hooker claim that accordingly, 'all the areas of design that involve any serious form of innovation against client briefs' – typical, for example, in architecture – are characterized by wicked problems as 'planning is a species of design' [2]. Moreover, architectural design for a sustainable future as well as heritage issues have been recognized as wicked problems in the context of a large EU-project about contemporary architectural education [25]. Healthcare architecture or management issues could also be mentioned here. Sustainability issues are met by considering many aspects from physical, ecological, technological, socio-political to economy-oriented ones, while heritage topics concerning preservation, conservation or restoration of historic environments are usually related to values and norms. There is also research done to improve education in order to prepare engineering students to address wicked problems within sustainability [26]. It would be worth discussing further if the theory and methods of addressing wicked problems could be recognized as a certain school of thought in architecture, similar to the sustainability school of thought in project management that is discussed by Gilbert Silvius [27]. He argues that the identity elements for any school of thought consist of a shared vision about the research paradigm, sets of concepts, community (academic and/or professional), and relevance of using certain methodologies, methods, and tools.

In the context of this essay, it is argued that the philosophical foundations of wicked problems include the issues of ontology of complex systems, the epistemology relevant to address different aspects of the system, various kinds of uncertainty, simultaneously as issues of axiology as values are often important drivers for research. Based on this, different dominant philosophical perspectives can be chosen (**Figure 3**). The approaches and methods to resolve complex design problems have been proposed by Johanna Lönngren as well as Coyan Tromp, plus others [26, 28], and include the following: systems thinking, action research, case-study methodology, collaborative strategies, participatory agendas, value-conscious strategies with a conscientious attention to values, methods to deal with uncertainty and identify knowledge gaps, and interdisciplinary or transdisciplinary approaches. Iterative approaches have been proposed by Raymond McCall and Janet Burge [29], and the proposition in this essay, which is to use research by design methodology to investigate wicked problems, is in line with the general idea of using iterative approaches.

Research by design: knowledge production and design processes

Research by design can be understood and interpreted in many ways, and the discussion of the concept is widespread and ongoing [30, 31]. In the current paper the definition in the European Association for Architectural Education's Research Charter is the primary reference:

> *In architecture, design is the essential feature. Any kind of inquiry in which design is a substantial constituent of the research process is referred to as research by design. In research by design, the architectural design process forms a pathway through which new insights, knowledge, practices or products come into being. It generates critical inquiry through design work.* [32]

The approach is alternatively called design research, practice-based research, or research through design, when it involved using design practice to generate academic knowledge. The latter has especially been adopted in the field of human computer interaction [33].

Within the field of architecture, the research by design approach is popular as a method of doing what could be called quasi-artistic research. However, it can play an important role in driving a working process especially while augmented with various other methods, and it can be seen as constituting the tool

integrating the scientific knowledge from different disciplines with the architect's implicit knowledge. Training in 'research by design' seems to be supported by Polanyi's concept of tacit knowledge relying on implicit acquisition of complex knowledge [8], and design can be treated as a system articulating tacit background knowledge revealed under feedback loops driven by perception. The actual object or artwork can be analysed from different perspectives while the holistic view is more complex.

The approaches and methods can be discussed in relation to the 'arrow diagram' (**Figure 4**), as modified from that in an essay by Halina Dunin-Woyseth and Fredrik Nilsson [34]. The methodologies used to produce and acquire knowledge (both implicit and explicit, by individuals and groups) extend between disciplinary scientific research (disciplinary 'Mode 1' knowledge acquisition matching scientific knowledge) and transdisciplinary creative practice (transdisciplinary 'Mode 2' knowledge acquisition as the result of dealing with practical questions and problems, and which is interdisciplinary, context-based with non-academics involved) [35, 36]. Transdisciplinary inquiry has better capacity to address complex problems referring to hybridization of knowledge which is of importance for architecture and urban planning [37]. 'Mode 1' and 'Mode 2' have a focus on ***episteme*** and ***techne*** respectively. It is also worth mention the 'Mode 3' of knowledge acquisition, in relation to existential questions and value-based human actions for a good life [28], which responds mostly to ***phronesis***.

Research by design is situated in the upper-right quadrant of the 'arrow diagram', responding to transdisciplinary research and creative practice, since it produces different kinds of knowledge including forms of practice-based knowledge. Knowledge generated by individuals and groups as a result of individual creativity, dialogue and participatory research is thus accumulated in practice over time. Eventually, explicit knowledge, tacit embodied knowledge, and not-yet embodied self-transcending knowledge acquired or produced through collaboration, can all be involved in that process.

Knowledge is created through interactions between tacit and explicit knowledge, rather than from tacit or explicit knowledge alone [38]. One of the interpretations of tacit knowledge, coming from the Vienna School, is provided by Gustavsson when he writes that 'the knowledge which exists on the basis of a long-term intimacy in the occupation, the tacit, can meet the knowledge produced in research, knowledge based upon assertions, and this leads to a development of knowledge in the practical occupation'. [5]

According to Ikujiro Nonaka and Hirotaka Takeuchi [39], there are four mechanisms of converting knowledge between the individual and the cooperating group:

- Socialisation through shared experience (tacit to new tacit)
- Externalisation through articulating (tacit to explicit)
- Combination through synthesis (explicit to new explicit)
- Internalisation through learning by doing (explicit triggers practice-related tacit)

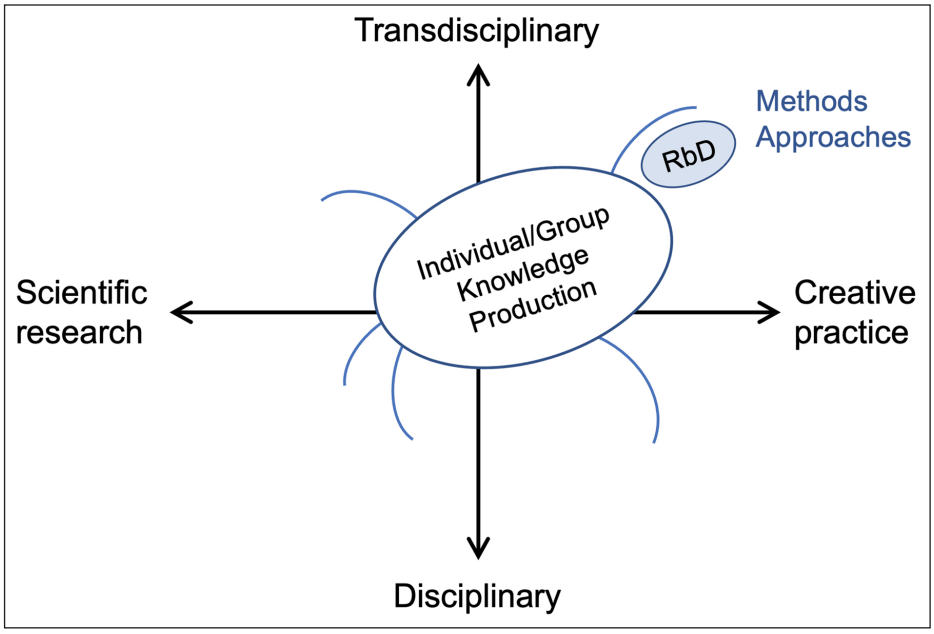


Figure 4: Four arrows of knowledge, as modified from Dunin-Woyseth and Nilsson [Diagram by the author].

Through the above conversion mechanisms, which can be understood as the modes of self-transcendence, new knowledge is generated, both tacit and explicit [40]. When it comes to the research by design approach, one of the challenges is the articulation, dissemination and transfer of the knowledge that is generated, and in this context, it is worth pointing out that the generation of explicit knowledge can reduce epistemological uncertainty.

Research occurs within a design process and has an impact on the designed object. Alain Findeli argues that 'behind every design question there lies a more fundamental research question that deals with the human habitability of the world' [13]. The knowledge of different kinds generated through design can also contribute to the development of theories, new insights and new concepts: 'Research by design generates critical inquiry through design work that may include realized projects, proposals, possible realities or alternatives' [32]. One of the many possible illustrations of the design process that is embedded in research by design inquiry can be tabulated as a sequence (**Figure 5**). The variety of these feedback loops shows the multiplicity of the design/research approaches that is dependent on the availability of empirical data, the stated problem, and finally the design research which has been specified. During the research process, designers constantly question the underlying criteria, and they redesign models in an investigative, creative and projective way. The process is guided throughout by different kinds of knowledge and critical reflection.

Research by design could be treated as an approach to tackle wicked problems of architectural design or urban planning, in which the mode of inquiry includes insights into future conditions which are uncertain and with consequences that are difficult to predict. When dealing with a wicked problem, extra loops may appear (shown in red in **Figure 5**). It means that the consequences of the wickedness in design or planning decisions, and their implementation, cause changes on a systemic level, and a post-intervention situation changes the way the problem is formulated. However, during the iterative process of a research by design trajectory, the level of epistemic uncertainty could be diminished according to the results which are being obtained.

A further diagram illustrates the possible role of research by design in knowledge production (**Figure 6**). Epistemic uncertainties of different kinds are decreasing during the iterative research/design process, triggering the production of scientific knowledge. Traditional sciences (basic sciences) can be seen as abstract and free from values and projections, and so their epistemic uncertainty is usually very small. Research in

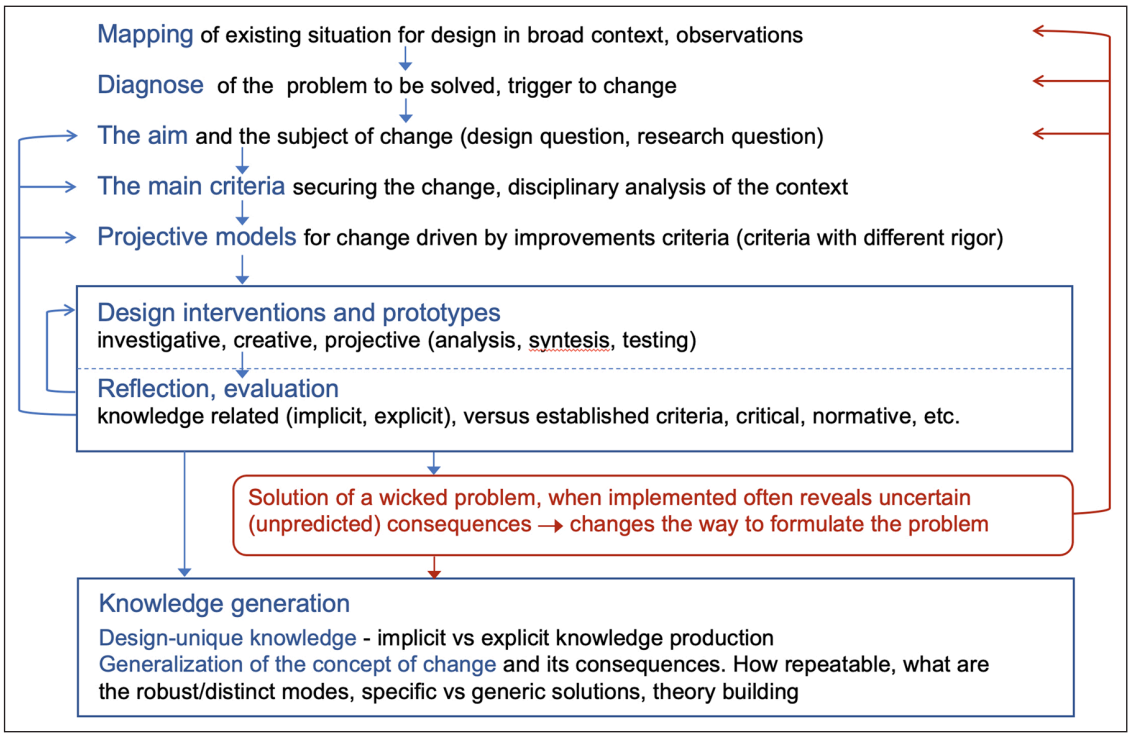


Figure 5: Research by design (in blue) methodology also when applied to a wicked problem (in red) [Diagram by the author].

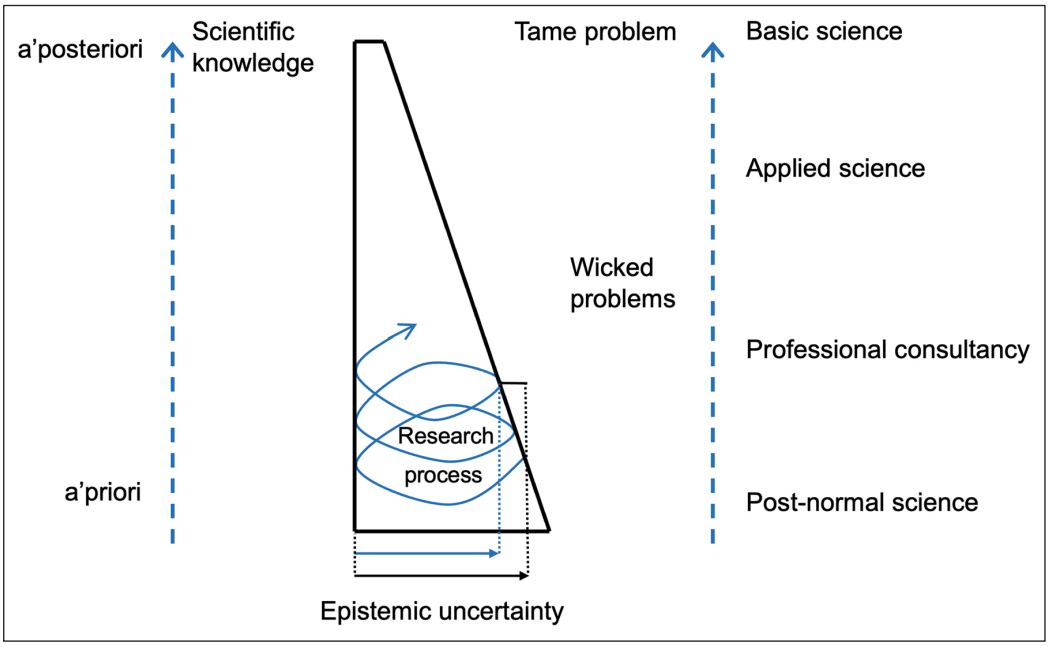


Figure 6: Knowledge production within the research by design process [Diagram by the author].

architecture, regarded as an applied science focusing on solving practical issues, is often projective and value-guided, resulting in a higher degree of uncertainty. This relates to the different types of research problems requiring different research methodologies, starting from logical or algorithmic problems to wicked case analysis, design problems and eventually so-called dilemmas that reconcile a complex, non-predictive decision with no solution [41]. In the case of wicked problems, epistemic uncertainty increases because of the ontological uncertainty and complexity. At the level of professional consultancy, design decisions are mainly implicit, and the base for decisions often unarticulated. Epistemic uncertainty is the highest in the case of so-called dilemmas to be approached by post-normal science [42]. Here, none of the traditional methodologies are effective and the problem/dilemma can only be evaluated as a dialogue that takes place within an extended peer community.

Research by design and wicked problems

In this essay, it is argued that the ontology of wicked problems refers to complex systems, and that research by design could be seen as a methodological approach to investigate or deal with wicked problems. According to McCall and Burge [29], the terms planning and design are used interchangeably by Rittel and Webber [1], both recognized as practices or fields consisting of wicked problems. Iterative approaches to design and planning that involve trans-disciplinarily knowledge production through sharing, collaboration, and co-creation give opportunities to resolve the problem (and generate specific knowledge) in better ways, while also giving input to the relevant theories (generic knowledge) and influencing their development. The ontology of complex systems, and the theory of wicked problems, are hence relevant for many architectural research applications. However, one must be aware of the ambiguity of these concepts, research by design included. As stated by Claes Andersson and Petter Törnberg [43], the concepts of complexity or wickedness suffer from lack of shared understanding that varies with the researcher's background, and it would be interesting to use the systematic meta-theoretical support that they present to approach complexity and wickedness in policy and design. The methodological approach for complex problems can employ a hybrid mode of inquiry [44] that is characterized by using mixed methods [45]. This is common when taking a pragmatic perspective on a research problem, whereby multiple methods from various disciplinary traditions can be incorporated in one study to resolve the research question in a good way (as in the perspectives of advocacy/participatory or critical realism).

To explain this point further, two PhD theses carried out at the Graduate School of Architecture at Chalmers University of Technology by Nils Björling and Joanna Gregorowicz-Kipszak can be briefly discussed as examples of how the role of research by design can play out in the context of wicked problems of urban planning and design, and also to unfold the systems of inquiry referred to by those PhD candidates when writing their theses.

The PhD thesis on 'Fragile Urban Landscapes: Planning methods to open spatial lock-ins of urbanization', by Nils Björling [4], focusses on the development of planning methods to break the reproduction of spatial lock-ins and to support renegotiations of the urban landscape (architectural thinking) through sketching, model making, theoretical reflection and suggestions of strategies in practice. In his doctoral project, a design-based research process was carried out whereby participation in planning practices enabled an active interplay between practice and theory (**Figure 7**). The ontology of the objective reality was in the thesis represented by assemblages called 'urban ecologies' (i.e. social, mental, and environmental) to describe the productive capacity of the city's landscape as a complex system. His research was driven by values in the form of design against fragility – a consequence of urban domination over regional areas in terms of resources. The use of a research by design methodology was obvious already from the beginning as the project, having been formulated by Björling (a former practicing architect), was based on his experience from previous design work. This can be defined as an advocacy/participatory paradigm, which can be seen as relevant when one is advocating for people from marginalized peripherical areas and calling for collaboration between the representatives of the groups from such areas and the researcher during the project. The iterative cyclic approach to design practice when fragile urban landscapes were renegotiated and reconfigured to open spatial lock-ins (**Figure 8**) has revealed one of the faces of the research by design approach used in the thesis.

Conversely, the PhD thesis on 'Rethinking Social Impact Assessment through Urban Design: Towards designerly evaluation with a socio-form approach' [3], by Joanna Gregorowicz-Kipszak, emphasizes a critical urban design perspective on the concepts and practices of social impact assessment (SIA) and draws attention to a designerly mode of evaluation (**Figure 9**). Her research project was intended to formulate the theoretical foundations for designerly strategies to develop social sustainability tools, as a complement to their contemporary employment for evaluations in urban planning and design practice. Introducing the concept of a socio-form approach indicates the use of an interdisciplinary research approach which integrates social

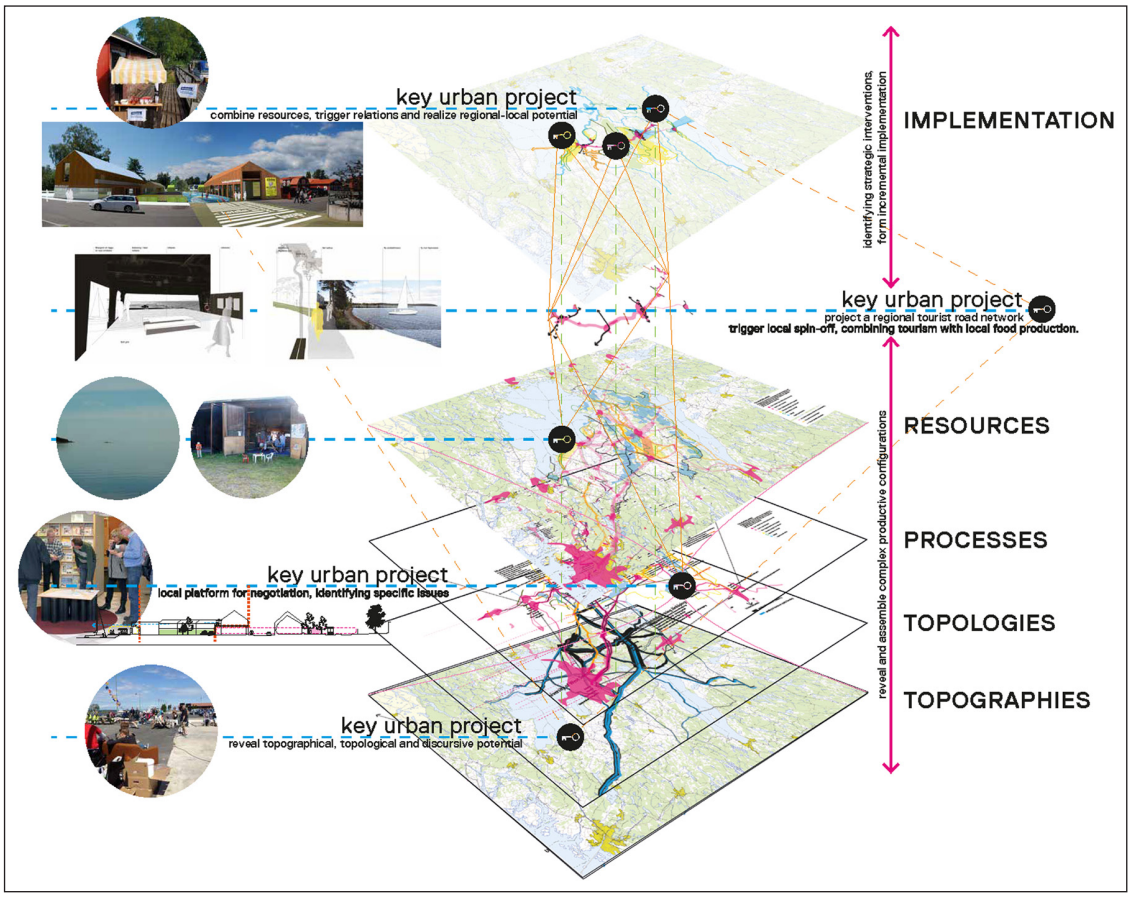


Figure 7: Urban projects as links between scales and themes to project and assemble urban ecologies [Courtesy of Nils Björling].

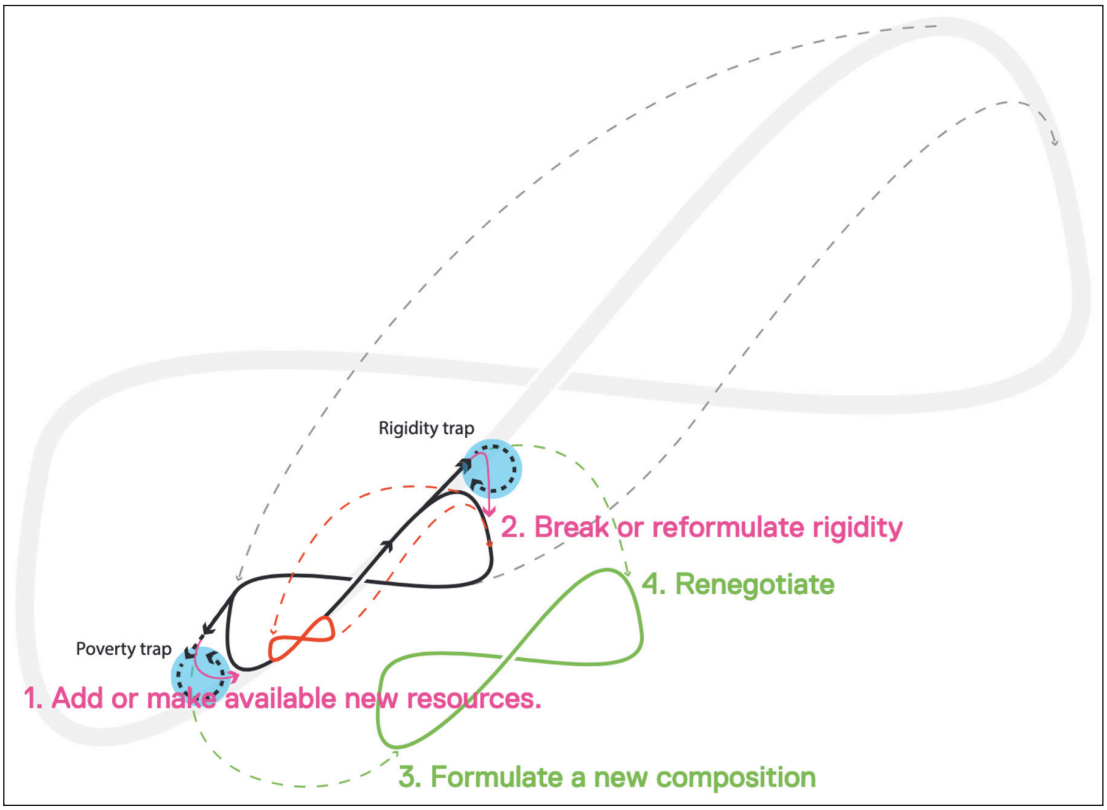


Figure 8: Four alternatives that in combination can open spatial lock-ins [Courtesy of Nils Björling].

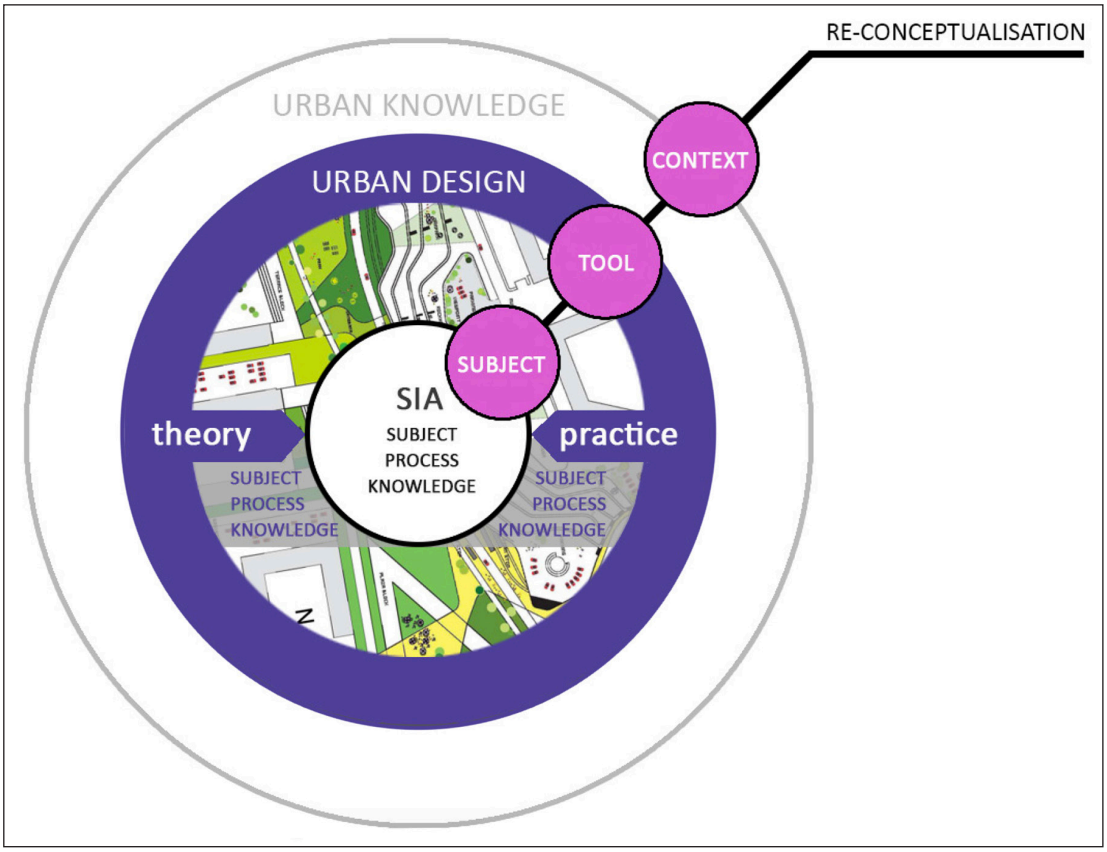


Figure 9: Reconceptualization of Social Impact Assessment through urban design within the context of urban knowledge production [Courtesy of Joanna Gregorowicz-Kipszak].

aspects with spatial/physical ones through power relationships (within a complex system) (**Figure 10**). The research in Gregorowicz-Kipszak's thesis concerns the ontological, epistemological, and methodological aspects of designing urban space within a socio-form approach. The research question in her thesis is subdivided into three parts: 1) perspectives on spatiality, urban space and its social aspects; 2) diversity of component activities, planning levels, and spatial scales that constitute urban design; and 3) wide range of stakeholders and types of knowledge involved in the design of urban space.

The research by design methodology in her thesis includes designerly approaches, such as screening, imaging, presenting, and testing, while also including studies of source literature and documents, participation in knowledge arenas, physical modelling, and focus group workshops (**Figure 11**). The study could also be seen as an argument for not trying to formulate just one strong way of social impact assessment, but instead pursuing constant discussion to give space for critical reflection and to encourage an awareness of designerly approaches in use. The use of research by design methodologies was not obvious at the beginning of the PhD project, but appeared later in the research process, retrospectively helping to realize and to articulate that she was somewhat unconsciously using the trained capacities and tools of a designer as well as of a researcher. As Gregorowicz-Kipszak writes: 'Research through design of an *approach to the design of urban space (socio-form)* was used as a method for advancing SIA, where design consists of a series of iterative activities applied on a local urban redevelopment case in Gothenburg' [3]. The research paradigm of this PhD study was defined by Sanda Lenzholzer et al. [46] as a constructivist one – according to the classification method offered by John Creswell and Viki Plano Clark [45] – and indeed

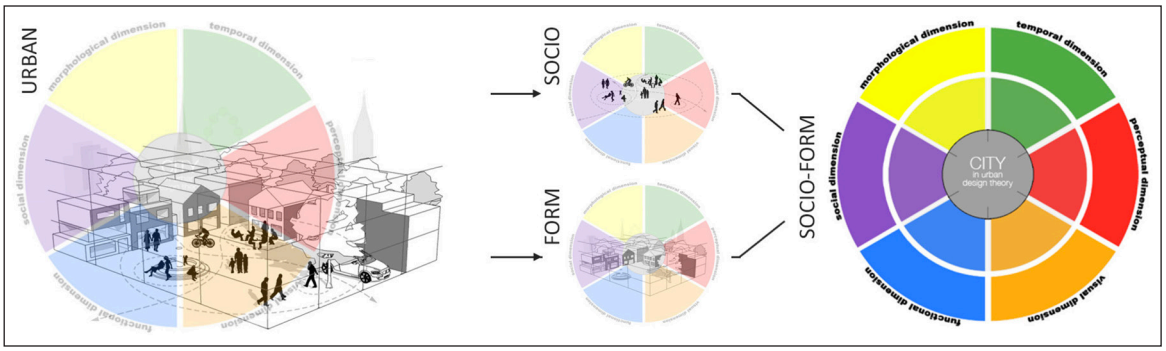


Figure 10: The design tactic of physical modelling in the socio-form approach [Courtesy of Joanna Gregorowicz-Kipszak].

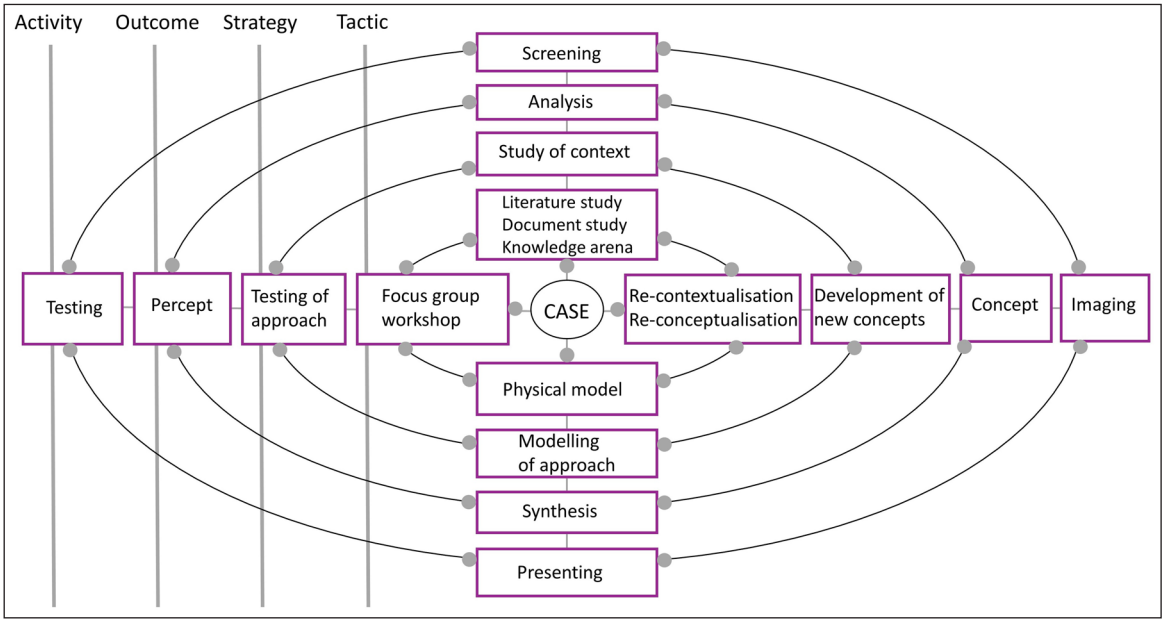


Figure 11: Research trough design: a dynamic representation of the applied methodological framework [Courtesy of Joanna Gregorowicz-Kipszak].

Gregorowicz-Kipszak's thesis has been listed as one of 40 examples of how to in a scholarly sense apply research through design methodology in the field of landscape architecture, out of 200 doctoral theses that were analysed. However, Gregorowicz-Kipszak's thesis mentions rather that her worldview is actually intended as (post)constructivist [46], and equally her research problematic could be seen from the perspective of critical realism, in that it acknowledges objective reality, interdisciplinary integration, and the independent causal powers of actions by agents (planners), prevailing social conditions, and spatial urban structures [47]. What this shows is that this study can be analysed through different focuses, and so it would require a far deeper understanding of Gregorowicz-Kipszak's research objectives, research processes and research outcomes to make a definitive statement. It could also indicate that a paradigmatic shift took place during the research process that is perhaps not reflected upon or presented explicitly. Anyway, clearly it is not easy to assign a dominant philosophical perspective when there is no neat classification of different philosophical stances.

These two examples of PhD studies which use research by design methodologies, albeit only briefly presented here, both seek to develop theoretical and methodological frameworks to solve the problems of urban design and planning in different ways. The viewpoints expressed in the theses tend to be holistic, setting the analysed system within a broader context (physical and social), and their proposed ontologies represent complex systems wherein different structures and agents are placed in relationships across the scales and issues involved. The theses show that it is rather straightforward to apply the theory of wicked problems to urban design and planning, even though the authors do not say this so explicitly. Drawing on the theory of wicked problems could open further discussions about the uncertainty and complexity of whichever system is under consideration.

Hence, both the PhD theses are inherently addressing the wicked problems of urban design and planning. The active role for design makes them close to action research, with participatory inquiry promoting a diversity of possible solutions in which a well-defined social goal is a strong driving force, as seen in Björling's thesis. Gregorowicz-Kipszak uses the research by design approach to refine the theory of socio-form approach, as well as to create a method/tool to use continuously throughout the design process to discuss the social consequences of design proposals by referring to knowledge about the context. Her research began by joining a pilot project and observing the design process from an academic perspective, which in turn favoured a conceptual structure for her thesis that relied separately on academic insight (represented by her as author of the PhD thesis) and the professional knowledge of the participants in the pilot project.

Generally, closer analysis of the sources of uncertainty within different kinds of knowledge could guide researchers towards more complete structures when they are designing their research project, using this analysis to judge how to deal with uncertainty under different systems of inquiry. Uncertainty is an immanent property of projective ideas typical for architectural research. All retroactive analysis of observations or results of experiment can be treated with scientific rigour, whereas prior strategies/designs are more challenging since they are built upon the epistemic uncertainty of different factors, thus letting implicit knowledge contribute. Once the solution has been worked out, through research by design methods, the uncertainty level is diminished, and some generalization of the results can be produced (**Figure 5**). And when the solution is implemented, new knowledge can be produced through a critical analysis of the consequences unless the system has behaved in a non-linear fashion and/or has run off track.

The reliability of this generated knowledge depends on the level of epistemic uncertainty of the research outcome. Furthermore, uncertainties in understanding the research issue or undertaking the processes of observation, experimenting, modelling, prototyping etc., also contribute to this epistemic uncertainty. When the design/research objective is set within a complex system that included emergent qualities of social or cultural dynamics, the uncertainty level becomes even higher. This can be lowered by using the research by design approach, yet one must still expect a significant level of so-called completeness uncertainty which reflects unanalysed aspects in the projected outcome.

From my experience of being active for many years in the doctoral courses for PhD students within ResArc – the Swedish Research School in Architecture, a cooperation between the four schools of architecture in Sweden [48] – I have observed a need to elaborate upon and map the scientific landscape in which the PhD researchers can clearly position themselves in relation to established research paradigms and worldviews. A basic understanding of one's philosophical stance and its epistemological consequences is important when designing the research project, and for developing research rigour, and it would be beneficial for those reading the final thesis to see the adopted research paradigm being formulated and discussed by the author. Doing so would enhance the clarity of the PhD study.

Conclusions

All research, and perhaps especially architectural research, calls for the need to develop and maintain academic rigour. Since the epistemic uncertainty of wicked problems is relatively high, applying the methodology of research by design (to such problems in particular) as an appropriate element in the chosen system of inquiry will support the scholarly legitimacy of the final thesis. The more specific conclusions of this essay can be summarised as follows:

- It is proposed that the philosophical underpinnings, theoretical background and methodology which form a research paradigm should become expected ingredients of any PhD thesis in architecture, including those using the research by design approach. Hence, there is a clear need for doctoral students to discuss the scientific landscape in which they position themselves in relation to research paradigms and worldviews.
- As PhD students in architecture often work with value-driven challenges, besides the ontological and epistemological perspectives of the research project, axiological assumptions should be discussed and if needed integrated into a consistent research paradigm.
- Wicked problems are common in architectural research, responding often to interdisciplinary or transdisciplinary challenges. Hence, they can form ontological classes of problems to be modelled using complex systems, and the relevant theories and methodologies can also be further developed.
- The relevance of research by design approaches to investigate wicked problems is now widely acknowledged, and so there are today many PhD theses that both address the wicked problems of urban design/planning and successfully use the research by design approach to deal with them.
- Being familiar with the theory of wicked problems gives students a chance to recognize and understand aspects such as complexity, uncertainty etc, and from there to choose the proper research paradigm accordingly and to discuss the specific place which the generated knowledge will sit in relation to scientific knowledge.
- The argument being made here is that for research by design approaches the generation of knowledge can help reduce epistemic uncertainty.

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Competing Interests

The author has no competing interests to declare.

References

1. **Rittel HWJ, Webber M.** Dilemmas in a General Theory of Planning. *Policy Sciences*. 1973; 4: 155–169. DOI: https://doi.org/10.1007/BF01405730
2. **Farrell R, Hooker C.** Design, science, and wicked problems. *Design Studies*. 2013; 34(6): 681–705. DOI: https://doi.org/10.1016/j.destud.2013.05.001
3. **Gregorowicz-Kipszak J.** Rethinking social impact assessment through urban design: Towards designerly evaluation with a socio-form approach. PhD thesis, Department of Architecture, Chalmers University of Technology; 2015.
4. **Björling N.** Sköra stadslandskap: Planeringsmetoder för att öppna urbaniseringens rumsliga inlåsningar. PhD thesis, Department of Architecture, Chalmers University of Technology; 2016.
5. **Gustavsson B.** The many faces of knowledge. Available at: http://media.skanesmoderna.se/2016/04/The-many-faces-of-knowledge-Bernt-G-Microsoft-Word-dokument.pdf.
6. **Gustavsson B.** *Kunskapsfilosofi: tre kunskapsformer i historisk belysning (Philosophy of Knowledge: Three Forms of Knowledge in a Historical Perspective)*. Wahlström & Widstrand; 2000.
7. **Popper K.** *Objective Knowledge: An Evolutionary Approach*. Clarendon Press; 1972.

8. **Polanyi M.** *The Tacit Dimension*. Routledge & Kegan Paul; 1966.
9. **Kuhn TS.** *The Structure of Scientific Revolutions (2nd Edition)*. University of Chicago Press; 1970: 43–51.
10. **Groat L, Wang D.** *Architectural Research Methods*. John Wiley & Sons; 2013.
11. **Easterby-Smith M, Torpe R, Lowe A.** *Management Research*. Sage Publications; 2002.
12. *Stanford Encyclopedia of Philosophy*. Available at: https://plato.stanford.edu/entries/architecture/#Ont.
13. **Findeli A.** The myth of the design androgyne. In: Joost G, Bredies K, Christensen M, Conradi F, Unteidig A (eds.) *Design as Research: Positions, Arguments, Perspectives*. Birkhäuser. 2016; 28–34. DOI: https://doi.org/10.1515/9783035607383-005
14. **Creswell J.** The selection of a research design. In: Creswell J (ed.) *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. Sage Publications. 2008; 3–21.
15. **Erckrath C, Andersson C, Kaps V.** Curricular Culture in PhD by Design: Perspectives gained from the Aarhus School of Architecture. *ARENA Journal of Architectural Research*. DOI: https://doi.org/10.5334/ajar.290
16. **Dyrssen C.** Navigating in heterogeneity: Architectural thinking and art-based research. In: Biggs M, Karlsson H (eds.) *The Routledge Companion to Research in the Arts*. Routledge. 2011; 223–239.
17. **Luck R.** Design research, architectural research, architectural design research: An argument on disciplinarily and identity. *Design Studies*. 2019; 65: 152–166. DOI: https://doi.org/10.1016/j.destud.2019.11.001
18. Chalmers University of Technology. Syllabus of the Graduate School in Architecture. Available at: http://www.chalmers.se/en/research/doctoral-programmes/graduate-schools/Pages/Architecture.aspx.
19. **Hansson OS.** What is technological science? *Studies in History and Philosophy of Science*. 2007; 38(3): 523–527. DOI: https://doi.org/10.1016/j.shpsa.2007.06.003
20. **Mo L.** *Philosophy of science for architects*. NTNU Norges teknisk-naturvitenskapelige universitet; 2001.
21. **Termeer CJAM, Dewulf A, Biesbroek R.** A critical assessment of the wicked problem concept: relevance and usefulness for policy science and practice. *Policy and Society*. 2019; 38(2): 167–179. DOI: https://doi.org/10.1080/14494035.2019.1617971
22. **Lönngren J, Svanström M.** Systems thinking for dealing with wicked sustainability problems: Beyond functionalist approaches. In: Leal Filho W, Nesbit S (eds.) *New Developments in Engineering Education for Sustainable Development (World Sustainability Series)*. Springer; 2016. DOI: https://doi.org/10.1007/978-3-319-32933-8_14
23. **Pietrzyk K.** A systemic approach to moisture problems in buildings for mould safety modelling. *Building and Environment*. 2015; 86: 50–60. DOI: https://doi.org/10.1016/j.buildenv.2014.12.013
24. **Pietrzyk K.** Bioclimatic tools for sustainable design: uncertainty perspective. *Proceedings of the 7th International Building Physics Conference IBPC18 Healthy, Intelligent and Resilient Buildings and Urban Environments*. Syracuse, USA. 2018; 511–516. DOI: https://doi.org/10.14305/ibpc.2018.gb-1.02
25. **Cabera I, Fausto I.** Confronting wicked problems: Adapting architectural education to the new situation in Europe. Final report of Erasmus+ project; 2015.
26. **Lönngren J.** *Wicked problems in engineering education*. PhD thesis. Department of Communication and Learning in Science, Chalmers University of Technology; 2017.
27. **Silvius G.** Sustainability as a new school of thought in project management. *Journal of Cleaner Production*. 2017; 166: 1479–1493. DOI: https://doi.org/10.1016/j.jclepro.2017.08.121
28. **Tromp C.** *Wicked Philosophy: Philosophy of Science and Vision Development for Complex Problems*. Amsterdam University Press; 2018. DOI: https://doi.org/10.1515/9789048541096
29. **McCall R, Burge J.** Untangling wicked problems. *Artificial Intelligence for Engineering Design, Analysis and Manufacturing*. 2016; 30: 200–210. DOI: https://doi.org/10.1017/S089006041600007X
30. **Verbeke J.** This is Research by Design. In: Fraser M (ed.) *Design Research in Architecture: An Overview*. Farnham: Ashgate. 2013; 137–159. DOI: https://doi.org/10.4324/9781315258126-8
31. **Van Reusel H, Michels C, Schoonjans Y.** Dissecting the Archipelago: PhD by Design Concepts in the Fields of Architecture and Urban Design. *ARENA Journal of Architectural Research*. 2021; 6(1). DOI: https://doi.org/10.5334/ajar.255
32. **European Association for Architectural Education.** EAAE Charter on Architectural Research; 2012. Available at: http://www.eaae.be/about/statutes-and-policypapers/eaae-charter-architectural-research/.
33. **Zimmerman J, Forlizzi J, Evenson S.** Research Through Design as a Method for Interaction Design Research in HCI. *CHI Proceedings, Design Theory*. San Jose, California USA; 2007. DOI: https://doi.org/10.1145/1240624.1240704
34. **Dunin-Woyseth H, Nilsson F.** Some notes on practice-based architectural design research: Four 'arrows' of knowledge. In: *Reflections +7: Research Training Sessions*. 2008; 138–147.

35. **Gibbons M, Limoges C, Nowotny H, Schwartzman S, Scott P, Trow M.** *The New Production of Knowledge: The Dynamics of Science and Research in Contemporary Societies*. Sage Publications; 1994.
36. **Dunin-Woyseth H, Nilsson F.** Building (Trans)Disciplinary Architectural Research: Introducing Mode 1 and Mode 2 to Design Practitioners. In: Doucet I, Janssens N (eds.) *Transdisciplinary Knowledge Production in Architecture and Urbanism*. Springer Science+Business Media. 2011; 79–96. DOI: https://doi.org/10.1007/978-94-007-0104-5_6
37. **Doucet I, Janssens N.** Transdisciplinarity, the hybridization of knowledge production and space-related research. Editorial. In: Doucet I, Janssens N (eds.) *Transdisciplinary Knowledge Production in Architecture and Urbanism*. Springer Science+Business Media. 2011; 1–14. DOI: https://doi.org/10.1007/978-94-007-0104-5_1
38. **Nonaka I, Toyama R, Konno N.** SECI, ba and leadership: A unified model of dynamic knowledge creation. *Long Range Planning*. 2000; 33: 5–34. DOI: https://doi.org/10.1016/S0024-6301(99)00115-6
39. **Nonaka I, Takeuchi H.** *The Knowledge Creating Company: How Japanese Companies Manage the Dynamics of Innovation*. Oxford University Press; 1995. DOI: https://doi.org/10.1016/0024-6301(96)81509-3
40. **Nonaka I, Konno N.** The concept of 'ba': Building a foundation for knowledge creation. *California Management Review*. 1998; 40(3): 40–54. DOI: https://doi.org/10.2307/41165942
41. **Jonassen DH.** Toward a design theory of problem solving. *ETR&D*. 2000; 48(4): 63–85. DOI: https://doi.org/10.1007/BF02300500
42. **Funtowicz SO, Ravetz JR.** Science for the post-normal age. *Futures*. 1993; 25(7): 739–755. DOI: https://doi.org/10.1016/0016-3287(93)90022-L
43. **Andersson C, Törnberg P.** Wickedness and the anatomy of complexity. *Futures*. 2018; 95: 118–138. DOI: https://doi.org/10.1016/j.futures.2017.11.001
44. **Zboinska MA.** Architectural Research in Hybrid Mode: Combining Diverse Methods within Design-Based Architectural Research Inquiry. *ARENA Journal of Architectural Research*. DOI: https://doi.org/10.5334/ajar.291
45. **Creswell JW, Plano Clark VL.** *Designing and Conducting Mixed Methods Research*. Sage Publications; 2011.
46. **Lenzholzer S, Nijhuis S, Cortesäo J.** RTD in landscape architecture: A first state of the art. In: Storni C, Leahy K, McMahon M, Lloyd P, Bohemia E (eds.) *Design as a Catalyst for Change. DRS International Conference 2018*. University of Limerick, Ireland; 2018. DOI: https://doi.org/10.21606/drs.2018.293
47. **Naess P.** Critical realism, urban planning and urban research. *European Planning Studies*. 2015; 23(6): 1228–1244. DOI: https://doi.org/10.1080/09654313.2014.994091
48. ResArc (Swedish Research School in Architecture). Available at: https://resarc.se/.

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